NITIN SN

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SUMMARY

MCA candidate (graduating 2027) with hands-on full-stack experience across React, Node.js/Express, and PostgreSQL, plus applied AI/ML in Python (FastAPI, scikit-learn). Built and deployed an ML-powered algorithm classifier and a full-stack booking platform. Strong foundation in data structures, algorithms, operating systems, DBMS, and computer networks.

TECHNICAL SKILLS

- Languages: Python, Java, JavaScript, TypeScript
- Web & Backend: React.js, Node.js, Express.js, FastAPI, REST API design, Tailwind CSS
- AI/ML & Data: scikit-learn, Pandas, Matplotlib, Random Forest, feature engineering
- Databases: MySQL, PostgreSQL, SQL, Database Design, Supabase
- Core CS: Object-Oriented Programming, Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks
- DevOps & Version Control: Git, GitHub; Docker, Kubernetes (coursework and self-study)
- Cloud & Tools: AWS (EC2, IAM, S3, Auto Scaling), Vercel, Render, Linux, VS Code

PROJECTS

AlgoLens — AI-Powered Algorithm Visualizer & Classifier | *React, TypeScript, Python, FastAPI, scikit-learn, Vercel, Render*

- Designed, built, and deployed an AI-powered algorithm visualizer end-to-end: React/TypeScript frontend on Vercel, Python FastAPI backend on Render. Live demo: algo-lens-viz.vercel.app
- Engineered a 54-feature AST-based extractor to parse code structure and represent it for machine-learning classification.
- Trained a Random Forest on 512 generated samples (86 unique structural patterns) across 17 algorithm categories: 74% leave-one-out accuracy on deduplicated patterns and 10/10 on independent programs; paired with a rule-based time/space complexity estimator.
- Integrated the classifier into the React frontend through a validated server-side function with typed error handling; Python is classified by the ML backend, while JavaScript, Java and backend failures fall back to a local AST/tokenizer analyzer.

PlayArena — Playground Booking System | *React, Node.js, Express, PostgreSQL, Supabase, Expo*

- Designed and built a full-stack sports playground booking platform for mobile (Expo) and web, from slot discovery to booking management, with real-time slot availability via Supabase Realtime.
- Fixed a double-booking race condition by moving booking into a Postgres function (Supabase RPC) that locks the slot row and enforces capacity in one transaction; a 20-parallel-request test accepted 16-17 bookings on a capacity-1 slot before the fix and exactly 1 after.
- Implemented role-based access control for players and venue owners with secure session handling.
- Built a venue-owner analytics dashboard from user feedback and ratings data.

EXPERIENCE

Rubixe AI — Software Development Intern Mar 2025 – Apr 2025

- Built a Python/Tkinter desktop app on the Spoonacular API: diet filters (vegan, vegetarian, gluten-free, etc.) narrow the recipe list, and selecting a recipe loads its image and instructions with pie and bar charts of its nutrients (Pandas, Matplotlib).
- Used Python multithreading to run Spoonacular API calls off the GUI thread, keeping the Tkinter app responsive.
- Built modular Python and Pandas scripts to ingest, validate, and transform nested REST API JSON responses into structured, analysis-ready datasets.

EDUCATION

Master of Computer Applications (MCA) 2025 – 2027
R.V. College of Engineering, Bengaluru — CGPA: 8.76

Bachelor of Computer Applications (BCA) 2022 – 2025
SSMRV College, Bengaluru — CGPA: 8.56

CERTIFICATIONS

- freeCodeCamp — Responsive Web Design Certification
- Udemy — AWS Fundamentals for Beginners